

# Orthogonality in Additive GPs for Interpretability

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IIB Technical Milestone Report

## Summary

Gaussian Process (GP) models are celebrated for their non-parametric flexibility and ability to provide principled uncertainty estimates. However, in high-dimensional settings, they face two critical challenges: scalability and interpretability. Additive Gaussian Processes (AGPs) partially address these by decomposing functions into interpretable components. Yet, the lack of unique decomposition, a phenomenon known as non-identifiability, limits their broader applicability. Orthogonal Additive Kernels (OAK) extend AGPs by introducing orthogonality constraints, ensuring unique, interpretable decompositions, which in turn affect the predictive power as discussed later.

This report examines the motivation for OAK, the theoretical underpinnings of additive and orthogonal kernels, and the practical challenges they address. We primarily discuss the implementation and evaluation of OAK on benchmark datasets and the development of Sobol-like indices for sensitivity analysis as this fits the future directions the best. It is worth noting that other tasks and achievements to date are not included but were useful in reaching this point (e.g. deriving valid conditions not explicated in the original literature analytically, exploring alternative research direction and implementing a codebase to allow for automated comparisons between kernels etc.) Finally, the report highlights future directions, including leveraging OAK in deep kernel learning and exploring advanced regularisation techniques parallels to OAK as a means of understanding the mechanism by which OAK sees discrepancies in performance.

## 1 Introduction and Research Objectives

Gaussian Process (GP) models have long been valued for their ability to model complex, non-linear relationships while providing uncertainty estimates. However, their adoption in high-dimensional data settings remains limited due to two interrelated challenges:

1. **Scalability:** Standard GP models face computational bottlenecks as the size and dimensionality of the data increase.
2. **Interpretability:** While GPs offer powerful predictive capabilities, understanding which features or interactions drive their predictions is often difficult, especially for high-dimensional data.

Additive Gaussian Processes (AGPs) [1] address the interpretability challenge by decomposing functions into sums of lower-dimensional components, isolating individual feature contributions and interactions. However, AGPs suffer from non-identifiability: multiple decompositions can yield the same function, leading to ambiguity in interpretation. Orthogonal Additive Kernels (OAK) [2], resolve this issue by enforcing orthogonality constraints on the kernel components. These constraints ensure that higher-order interactions do not interfere with lower-order terms, yielding unique and interpretable decompositions.

This project builds on the foundational work of AGPs and OAK, aiming to address the following research questions:

1. How can additive and orthogonal kernel structures be effectively developed and evaluated to isolate feature interactions and assess their relative performance?
2. How do Sobol-like indices quantify the contributions of different interaction orders in OAK, and what insights do they provide about model behaviour?
3. Why does OAK exhibit improved interpretability and changes in performance over standard additive kernels in empirical evaluations, and under what conditions are these improvements most significant?

By tackling these challenges, this work seeks to enhance the practical applicability of GPs in complex, high-dimensional settings, bridging the gap between interpretability and predictive performance. This area of research would be especially useful in domains such as Finance and Healthcare where predictive performance alone is not a sufficient metric, as understanding the roles of the individual factors involved is vital.

## 2 Theoretical Framework

### 2.1 Additive Gaussian Processes

Duvenaud et al. [1] introduced Additive Gaussian Processes (AGPs), which improves interpretability by decomposing functions as:

$$f(\mathbf{x}) = f_1(x_1) + f_2(x_2) + \dots + f_{12}(x_1, x_2) + \dots + f_{12\dots D}(x_1, x_2, \dots, x_D), \quad (1)$$

where  $\mathbf{x} \in \mathbb{R}^D$ , and  $x_i$  is its  $i^{th}$  component. This is achieved through a kernel constructed hierarchically:

$$k_{add_1}(\mathbf{x}, \mathbf{x}') = \sigma_1^2 \sum_{i=1}^D k_i(x_i, x'_i), \quad (2)$$

$$k_{add_2}(\mathbf{x}, \mathbf{x}') = \sigma_2^2 \sum_{i < j} k_i(x_i, x'_i) k_j(x_j, x'_j), \quad (3)$$

$$k_{add_n}(\mathbf{x}, \mathbf{x}') = \sigma_n^2 \sum_{i_1 < \dots < i_n} \prod_{d=1}^n k_{i_d}(x_{i_d}, x'_{i_d}). \quad (4)$$

Note that a  $0^{th}$  order kernel will be a constant kernel. Once a one-dimensional base kernel is denoted for each dimension, the full kernel can be formed by summing these terms. In this manner, AGPs improve interpretability by isolating the contributions of features and their interactions. By in turn analysing the contribution of each kernel, by directly getting the ratio of the overall variance that a single component is responsible for, we can interpret how each order of interactions has contributed in the sum. This is further discussed in the Sobol-like indices section.

### 2.2 The Identifiability Challenge

Despite their benefits, the original AGP framework suffers from a fundamental issue of *non-identifiability* [2]. This arises because multiple decompositions can yield the same overall function. For example, in a two-dimensional case with only 0th and 1st order terms:

$$f(x_1, x_2) = 1 + f_1(x_1) + f_2(x_2), \quad (5)$$

an equivalent, but less interpretable, decomposition could be:

$$f(x_1, x_2) = (1 + \Delta) + (f_1(x_1) - \Delta) + f_2(x_2). \quad (6)$$

This ambiguity stems from higher-order interactions interfering with lower-order terms, complicating the interpretation of the contributions from each feature or interaction. Without additional constraints, the framework allows for an infinite number of plausible solutions, undermining the goal of interpretability.

### 2.3 Orthogonal Additive Kernels (OAK)

To resolve the identifiability issue, Lu et al. [2] introduced orthogonality constraints, which enforce unique functional decompositions. For each feature  $x_i$ , the orthogonality condition is:

$$\int f_i(x_i)p(x_i)dx_i = 0, \quad \forall i, \quad (7)$$

where  $p(x_i)$  represents the probability density of  $x_i$ . This ensures that each component  $f_i$  is uncorrelated with the constant term, thereby preventing interference between different orders of interaction.

The base kernel  $\tilde{k}_i(x_i, x'_i)$  is modified to satisfy this constraint:

$$\tilde{k}_i(x_i, x'_i) = k_i(x_i, x'_i) - \int k_i(x_i, x_i)p(x_i)dx_i \left[ \int \int k_i(x_i, x'_i)p(x_i)p(x'_i)dx_i dx'_i \right]^{-1} \int k_i(x'_i, x_i)p(x_i)dx_i \quad (8)$$

This *constrained* kernel subtracts the expected contributions of  $k_i(x_i, x'_i)$  under  $p(x_i)$ , ensuring that the first-order components are orthogonal to the constant term and higher-order interactions do not interfere with lower-order terms. This formulation guarantees unique and interpretable decompositions in the AGP framework.

### 2.4 Sobol-like Indices

Sobol indices are a powerful tool for global sensitivity analysis, quantifying the contributions of individual features and their interactions to the variance of a model's output. For a function  $f(\mathbf{x})$ , the first-order Sobol index for feature  $x_i$  is defined as:

$$S_i = \frac{\text{Var}_{x_i}(\mathbb{E}[f(\mathbf{x}) \mid x_i])}{\text{Var}(f(\mathbf{x}))}, \quad (9)$$

which measures the proportion of output variance explained by  $x_i$  alone. Similarly, higher-order indices quantify the contributions of feature interactions:

$$S_{ij} = \frac{\text{Var}_{x_i, x_j}(\mathbb{E}[f(\mathbf{x}) \mid x_i, x_j]) - S_i - S_j}{\text{Var}(f(\mathbf{x}))}. \quad (10)$$

The total Sobol index for a feature  $x_i$  aggregates all contributions involving  $x_i$ , including its interactions:

$$S_{T_i} = S_i + \sum_{j \neq i} S_{ij} + \dots \quad (11)$$

In the context of OAK, orthogonality constraints simplify the computation of Sobol indices. Since each interaction order is isolated and uncorrelated with others, the variance contributions of individual terms can be directly attributed to specific features or interactions. This decomposition enables a clearer understanding of how features influence the model output.

## 3 Methods

For the purposes of this report, we focus on the autoMPG dataset, a widely used benchmark for regression tasks, while similar analyses were performed on other datasets such as Pima and Pumadyn. These were preprocessed identically to [1] and [2]. The autoMPG dataset contains

information on 398 automobiles produced between 1970 and 1978, including features such as horsepower, weight, acceleration, and miles per gallon, as well as the target variable of miles per gallon.

The models evaluated include Additive Gaussian Processes (AGPs) and Orthogonal Additive Kernels (OAK), which were implemented in Python using TensorFlow and GPflow. To ensure robust evaluation, we employed k-fold cross-validation (with  $k = 5$ ), splitting the data into training and test sets across folds with a 80:20 split for training and testing.

Key metrics used to assess model performance include Root Mean Squared Error (RMSE) for predictive accuracy and Negative Log-Likelihood (NLL) for evaluating probabilistic calibration. These were chosen to directly compare with the reported values in [2]. Sobol-like indices were computed to quantify the contributions of individual features and interaction orders to model variance, providing insight into model interpretability.

This methodological approach enables a comprehensive comparison of AGPs and OAK across datasets, while ensuring fair evaluation and interpretability assessment.

## 4 Results and discussion

### 4.1 Performance Evaluation

Figures 1 and 2 compare the Negative Log Likelihood (NLL) and Root Mean Squared Error (RMSE) for additive and orthogonal kernels on the autoMPG dataset. OAK is observed to have maintained competitive performance, even after the orthogonality constraint. However, it is worth noting that OAK has a worse NLL fit on average across the 5 folds. This is perhaps because, if we interpret OAK to be a constrained version of Additive, we can see that it would have reduced flexibility and a worse *data fit* term.

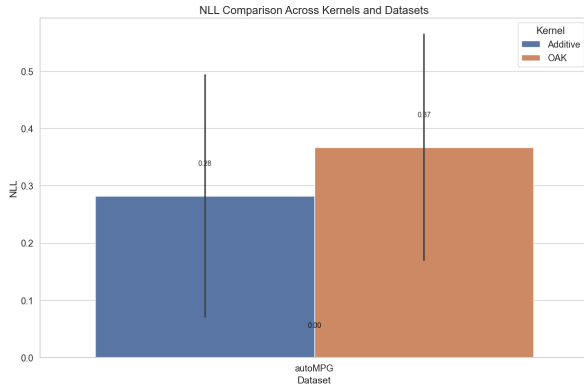


Figure 1: NLL comparison between Additive and OAK kernels on autoMPG dataset.

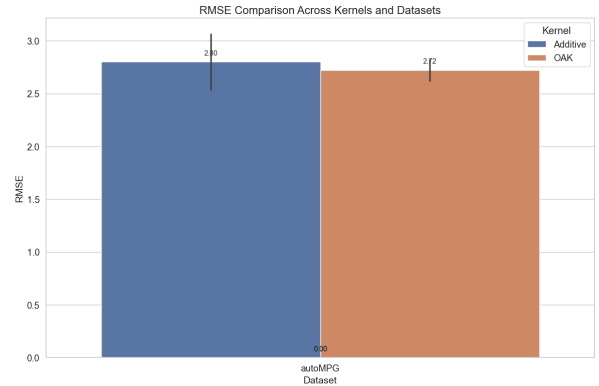


Figure 2: RMSE comparison between Additive and OAK kernels on autoMPG dataset.

### 4.2 Sobol Index Analysis

The Sobol-like indices in Figure 3 illustrate the variance contributions across interaction orders for the autoMPG dataset. For the OAK model, the vast majority of the variance contribution comes from the zeroth order (0.98), with a very small contribution from second-order interactions (0.02). This suggests that the OAK model is heavily reliant on the constant term, effectively capturing the overall mean trend in the data, with minimal influence from interactions between individual features like “cylinders,” “displacement,” or “weight.” The orthogonality constraints enforced in OAK lead to an interpretable decomposition where the majority of the predictive power is concentrated in the simpler, lower-order terms.

In contrast, the Additive Kernel model distributes variance contributions across multiple orders. Notably, the zeroth-order contribution is much smaller (0.0631), indicating a weaker

overall mean trend. First-order contributions dominate (0.6567), highlighting that individual features like “cylinders,” “displacement,” and “horsepower” are the primary drivers of predictive variance. Higher-order interactions, such as second-order (0.0266) and fourth-order (0.2140), also play a non-negligible role in explaining the variance. These contributions suggest that the Additive Kernel captures more complex dependencies between combinations of features, such as the interaction between “weight” and “acceleration” or “cylinders” and “year.”

This difference highlights the interpretability advantage of OAK. By concentrating variance in lower-order terms, OAK reduces the complexity of the functional decomposition, making it easier to attribute predictions to specific features or simpler interactions. On the other hand, the Additive Kernel allows for greater flexibility by distributing contributions across higher-order interactions but sacrifices interpretability due to the difficulty in disentangling these contributions.

For example, in the autoMPG dataset, understanding how “weight” or “year” independently influences predictions is more straightforward with OAK, as its predictions are less confounded by complex interactions. Meanwhile, the Additive Kernel model may capture subtle dependencies, such as the combined influence of “horsepower” and “acceleration,” but at the cost of interpretability. This makes OAK particularly well-suited for scenarios where interpretability is paramount, such as in applications requiring feature attribution or explanation of model outputs.

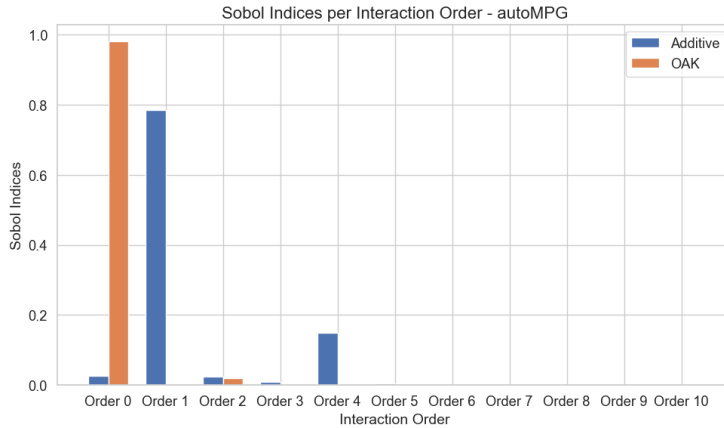


Figure 3: Sobol indices per interaction order for Additive and OAK kernels on autoMPG dataset. OAK shows greater contributions from lower-order terms, enhancing interpretability.

## 5 Future Works

### 5.1 Investigating Marginal Likelihood and Constraints in OAK Models

The constraints in OAK raise important questions about their impact on marginal likelihood optimisation. Does the additional constraint in OAK lead to a trade-off in flexibility as we suspect, potentially resulting in a worse marginal likelihood compared to unconstrained additive models analytically? Investigating this could clarify whether these constraints improve interpretability at the cost of predictive performance. Additionally, aligning optimisation with methods like those of Lu [2], who uses normalising flows to enforce the Gaussian measure assumption, or exploring alternatives to relax these requirements, could enhance both training and model performance.

### 5.2 Regularization in OAK Models

The observed performance of OAK models could be linked to overfitting, particularly in high-dimensional or sparse data scenarios. Dropout, a well-known regularisation technique in neural networks, has parallels with Gaussian Processes and may offer a practical solution. By acting as a Bayesian approximation [5], dropout could serve as a stochastic regulariser parallel for OAK models, mitigating overfitting while preserving interpretability. A rigorous comparison between

dropout and traditional priors used in additive GP frameworks could provide further insights. Exploring these connections might also reveal mechanisms underlying OAK’s ability to balance predictive accuracy and interpretability.

### 5.3 Advanced Sensitivity Analysis with Sobol-like Indices

Sobol-like indices quantify feature importance and interactions, but their potential for deeper analysis remains underexplored. Could the distribution of Sobol indices reveal insights into model generalisation? For instance, do models with concentrated variance contributions in lower-order terms generalise better? Investigating this relationship could enhance our understanding of how interpretability and predictive performance interplay. Furthermore, developing alternative metrics for feature contributions or extending Sobol indices to large-scale or hierarchical settings could further advance sensitivity analysis in OAK and additive GP frameworks.

### 5.4 Additive Kernels in Deep Kernel Learning (DKL)

Deep Kernel Learning (DKL) combines the flexibility of deep neural networks with the uncertainty quantification of Gaussian Processes. Incorporating additive and orthogonal kernels into DKL could serve as an automatic regulariser, balancing scalability and interpretability. Additive decomposition might allow for hierarchical feature interactions, improving the model’s ability to capture complex dependencies while maintaining clear attribution of feature contributions. Moreover, integrating OAK constraints could address limitations in modelling higher-order dependencies, offering a natural path to scalable, interpretable frameworks for large and intricate datasets.

## 6 Next steps

The immediate next steps will be expanding the code base to include classification based datasets too, and utilising synthetic datasets to explore the behaviour of OAK and additive better. For example, a synthetic dataset which  $N$  dimensions but an output only reliant on 1 dimensions should have the same behaviour for both models, since there are no interaction terms that are relevant. The risks for this term are mainly related to progress being hindered by scattered exploration about all aspects of OAK. To address this, I have created weekly aims and tasks which will be reviewed regularly with Prof. Carl Rasmussen to ensure the overall aim of finding the reason for the performance differences is found in a more focussed manner.

## References

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